disease seemed extremely mild. "People are now majorly recovering at home through telemedicine," says Dr Sushila Kataria, internal medicine specialist at Medanta - The Medicity in Gurgaon. It has bred a certain callousness towards the virus. "People are tending to forget that the elderly, or those with low immunity, are still at high risk. If one has symptoms, they must test, isolate and get treated."



DROPPED GUARD Maskless crowd at the Bankey Bihari temple in Vrindavan, June 1

Medical experts unanimously caution not to let our collective guard down in the belief that official figures are low or the disease is mild. "Every virus, like every other living organism, wants to survive," says Dr Rakesh Mishra, director of the Bengaluru-based Tata Institute for Genetics and Society. "There is no indication that the current lot of sub-variants is unusual, or will cause surges. But it does show that Covid is trying to survive. And there could still be an entirely new variant that could cause another surge."

A GUIDE TO THE GROWING OMICRON FAMILY

- BA.1 was the **ORIGINAL VARIANT**. Almost one new sub-lineage of it is being designated every day
- BA.1, BA.2, BA.3, BA.4, ba.5 were the **FIRST 5 DESCENDANTS OF THE ORIGINAL** Omicron ancestor. Each has bred further lineages with new mutations
- All subvariants are different but when they belong to a larger grouping (like Omicron), they aren't assigned Greek letters
- BA.2 is the **DOMINANT STRAIN WORLDWIDE** and has around 21 sub-lineages
- BA.2.12.1, BA.4 and BA.5 are the **3 SUB-LINEAGES RESEARCHERS ARE CONCERNED ABOUT GLOBALLY.** The first started a new wave in the US; the other two caused South Africa's fifth wave
- All three contain the **L452 MUTATION**, which allows the virus to infect both nasal and lung tissues, leading to more severe disease
- BA.1 had a threefold greater reproduction number compared to Delta. BA.2, which replaced BA.1 was 30% more transmissible, and **BA.2.12.1 IS 25% MORE TRANSMISSIBLE**
- Omicron also has a number of **RECOMBINANT VARIANTS**, or when two sub-variants combine in a host. These have so far not behaved differently from the parent variant
- Experts are closely monitoring the **XD VARIANT IN FRANCE AND XF IN THE UK**, which are a product of Delta and BA.1 combining. They could contain mutations that make them more severe as well as more transmissible

NO MORE GREEK LETTERS

A record 11 million Covid genomes have been uploaded on the GISAID platform since January 2020, compared to only 1.6 million variants of the influenza virus EpiFlu uploaded since May 2008. It does not tell us Covid is mutating faster than most viruses or whether researchers are simply uploading more data for it. A majority of Covid's mutations are mundane and are constantly ousted by newer, fitter variants. This will be till a variant emerges that is more superior in infecting humans, so much so that it replaces all other variants, infecting even those who have been afflicted with previous variants. Every such variant, however, may not be assigned a Greek letter of its own.

Typically, any variant attracting enough attention is submitted to a bioinformatics system called Pango, which then compares the data to hundreds of other samples, to ascertain whether the variant is a new brand of the Covid family tree or simply a sub-brand of an older one. For example, BA.1, BA.2, BA.3, BA.4 and BA.5 are the first five branches of the Omicron ancestor. BA.2.12.1 is the 12th lineage to branch off from BA.2, and then the first named branch on the 12th brand. Essentially, these sub-variants are fresh avatars but their mutations are still not different enough to qualify them as new brands and desrving of a separate Greek letter. Only when a variant escapes the immune system much more effectively than others or causes more severe disease or is much more transmissible does the WHO brand it as a 'variant of concern' and award it a Greek letter. So far, only five variants of Covid have met this criteria—Alpha, Beta,